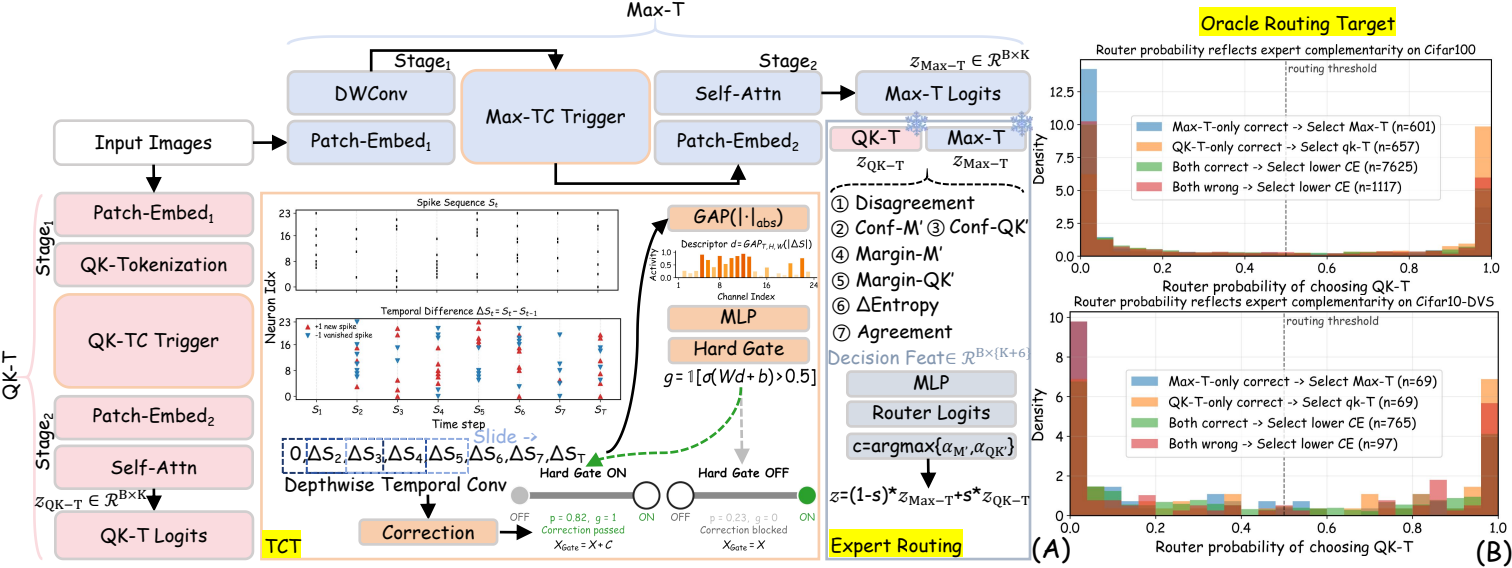

Max-T
Stage1
DWConv
Patch-Embed1
Max-TC Trigger
Stage2
Self-Attn
Patch-Embed2
Max-T Logits
Input Images
QK-T
Stage1
Patch-Embed1
QK-Tokenization
QK-TC Trigger
Stage2
Patch-Embed2
Self-Attn
QK-T Logits
Spike Sequence
Neuron Idx
Temporal Difference
+1 new spike
-1 vanished spike
Time step
GAP(|·|abs)
Descriptor
Activity
Channel Index
MLP
Hard Gate
Slide
Depthwise Temporal Conv
Hard Gate ON
Hard Gate OFF
Correction
OFF
ON
Correction passed
Correction blocked
TCT
QK-T
Max-T
① Disagreement
② Conf-M' ③ Conf-QK'
④ Margin-M'
⑤ Margin-QK'
⑥ ΔEntropy
⑦ Agreement
MLP
Router Logits
c=argmax{α,α}
Expert Routing
Oracle Routing Target
Router probability reflects expert complementarity on Cifar100
routing threshold
Max-T-only correct -> Select Max-T (n=601)
QK-T-only correct -> Select qk-T (n=657)
Both correct -> Select lower CE (n=7625)
Both wrong -> Select lower CE (n=1117)
Density
Router probability of choosing QK-T
Router probability reflects expert complementarity on Cifar10-DVS
Max-T-only correct -> Select Max-T (n=69)
QK-T-only correct -> Select qk-T (n=69)
Both correct -> Select lower CE (n=765)
Both wrong -> Select lower CE (n=97)
(A)
(B)